

SIEGFRIED BOZZA

Full-Stack Engineer · Web3 & AI Agents · Remote-first · Open to global relocation

Annecy, France · siegfried.bozza@protonmail.com · [GitHub](#) · [Portfolio](#) · [LinkedIn](#)

Target roles: Web3 full-stack · Smart-contract engineering · DeSci · Agentic backends · Web2 full-stack

PROFESSIONAL SUMMARY

M.Sc. Molecular Biology + **M.Eng.** Civil / Environmental Engineering — three peer-reviewed publications and roughly fifteen years across research and field engineering in **France**, **Canada**, and **Switzerland** — before transitioning to software in 2022. **Full-Stack Engineer** at Prismo.io (Next.js / TypeScript, CQRS, hexagonal architecture, Supabase, Vitest / Cypress). Four personal projects: three Web3 DApps on Base and Ethereum Sepolia (Solidity, Foundry, Chainlink, LangGraph, Inngest, EIP-712, event-driven CQRS) and one full-stack Web2 auction platform.

TECHNICAL SKILLS

Frontend & UI: React 19, Next.js (App Router), TypeScript, TanStack Query / Table, nuqs, Tailwind CSS, shadcn/ui, Zod, Recharts, React Hook Form, Lingui

Web3 & Smart Contracts: Solidity, Foundry (contracts, tests, deploy scripts), OpenZeppelin (ERC-20/721/1155), Chainlink (VRF v2.5, Automation, Price Feeds), Viem, Wagmi, EIP-712, Reown AppKit, IPFS, Alchemy, Tenderly

AI & Orchestration: LangGraph.js (agentic graphs, HITL, checkpointers), Inngest (durable execution), Gemini, Exa AI

Backend & Data: Node.js, Supabase (RPC, RLS, Realtime, Edge Functions, CRON), Neon Postgres, Drizzle ORM, CQRS, event-driven design, ports & adapters, Stripe API

Testing & DevOps: Foundry, Vitest, Cypress, Jest, GitHub Actions, Vercel, CI/CD, Git Flow

Languages: French (native) · English (fluent) · German (B1)

WORK EXPERIENCE

Prismo.io — Hybrid, France · Full-Stack Engineer · Aug 2024 – Present

- Real-time, data-heavy UI — TanStack Table data grids, server-side URL state (nuqs), Recharts KPI dashboards on shadcn/ui + Tailwind CSS; Next.js streaming + React Suspense; i18n (Lingui).
- CQRS + event-driven architecture — read / write separation via Supabase primitives (database triggers, RPC functions, typed client).
- Hexagonal architecture — domain logic decoupled from infrastructure via ports / adapters.
- Testing — Vitest unit tests + Cypress end-to-end on critical workflows; required CI checks before merge to main.
- AI-augmented delivery — agents in Cursor for planning and implementation; agent output reviewed under the same PR gates as human code.
- Agile — two-week sprints in Linear, peer code reviews on every PR, Git Flow branching.

Freelance — Remote, France & Indonesia · Full-Stack Developer · Jul 2023 – Aug 2024

- Shipped React / Next.js applications with Tailwind CSS, shadcn/ui, Framer Motion, MongoDB, and Jest.
- Contributed front-end features to DeXter-on-Radix (open-source DEX DApp) in an async distributed team.
- Worked remotely from Bali — comfortable with async communication and self-directed delivery across timezones.

Renuo AG — Zurich, Switzerland · Software Engineer Intern · Jul 2022 – Jun 2023

- Built and shipped full-stack features across multiple agency clients in Ruby on Rails, JavaScript / TypeScript, and React; RSpec, clean-code practices, agile delivery.

TECHNICAL PROJECTS

BioVerify — Agentic DeSci DApp · Mar–Apr 2026 · [GitHub](#) · [Article](#) · [Live](#)

- Peer review as a coordination game: authors submit research artifacts pinned to **IPFS** and stake ETH, a **LangGraph** submission agent screens plagiarism (Exa AI + Gemini), **Chainlink VRF v2.5** selects reviewers from the staked pool, and human verdicts settle on-chain through a LangGraph review agent (**HITL**, **EIP-712** signatures, on-chain transactions).
- Getter-less Solidity contract; **Alchemy** Notify webhooks (HMAC verified) project events into **Neon** Postgres (Drizzle) with optimistic concurrency; **Wagmi**, **Viem** WebSockets + **TanStack Query** drive the live UI.
- Durable orchestration: **Inngest** (step.run, retries) and LangGraph checkpointers carry multi-day human-in-the-loop review across serverless cold starts.
- BioVerify contract: 50 **Foundry** tests across 12 suites — 100% lines / statements / branches / functions; deployed on Base Sepolia and Ethereum Sepolia.
- Public **Telegram bot** mirrors contract state transitions across both testnets.

Bet2Gether — Chainlink-Settled Prediction Market · Oct–Nov 2025 · [GitHub](#) · [Live](#)

- Operator-free prediction market on Ethereum Sepolia: **Chainlink Automation** calls *performUpkeep* to resolve rounds at deadline without operator intervention; settlement reads **Chainlink Price Feeds**.
- Time-weighted payouts (earlier bets earn a larger share); CEI pattern and OpenZeppelin nonReentrant on claimReward.
- Chainlink VRF v2.5 for unpredictable ERC-1155 reward token IDs; a Tenderly Web3 Action bridges resolution events to a role-gated mint.

Forge — Atomic ERC-1155 Crafting · Oct–Nov 2025 · [GitHub](#) · [Live](#)

- Experiment in ERC-1155 crafting rules: rules contract (Forge) vs ledger (ERC-1155 FToken) with immutable ownership; composite IDs 3–6 minted via atomic burnBatch + mint in one transaction.
- Cooldown-gated basic mints (IDs 0–2); event-driven Next.js UI with Wagmi, **Alchemy WebSockets** and TanStack Query.
- Foundry tests — 100% coverage on project-owned contracts (lines / statements / branches / functions).

GavL — Next Auctions: Real-Time Auction Platform · Sep–Oct 2025 · [GitHub](#) · [Live](#)

- Full-stack Web2 auction platform on Next.js + **Supabase**, with i18n FR / EN / DE (Lingui).
- **Supabase** Realtime for live bids and notifications; a CRON job triggers an Edge Function for scheduled auction closure.
- Stripe Checkout with verified webhooks; service-role Supabase client safely bypasses RLS to update payment states asynchronously.
- URL-driven filters (nuqs), **TanStack** Table, Recharts dashboards, Jest tests

EARLIER CAREER (PRE-SOFTWARE)

KIBAG Gruppe · SERPOL · SITA Remediation (SUEZ) — France & Switzerland · Land Remediation Engineer / Project Manager · 2012–2020

- Led environmental fieldwork, soil-sampling operations, subcontractor coordination, and compliance reporting for industrial remediation sites.

University of British Columbia · INSERM — Canada & France · Molecular Biology Researcher · 2001–2009

- Genomic research in neural stem cells and epigenetics; co-author on three peer-reviewed publications ([J. Neurosci. Res.](#) · [Fungal Genet. Biol.](#) · [Dev. Biol.](#)).

EDUCATION

2025 — EVM Architecture & Smart Contract Security (core curriculum) — Metana (remote)

2021–2023 — M.Sc. Blockchain Technologies — University of Barcelona (remote)

2022 — Full-Stack Software Engineering Bootcamp — Le Wagon, Zurich

2012 — M.Eng. Civil & Environmental Engineering — École des Mines, France

2001 — M.Sc. Molecular Biology & Genetics — University of Montpellier, France

[GitHub](#) · [Portfolio](#) · [LinkedIn](#)